Appendix C: Uncertainty Budget Case Study: CMM Length Measurements

This appendix introduces the steps involved in performing a detailed uncertainty budget analysis for CMM length measurements. Related reading is provided for formulating uncertainty budgets in other measurement areas including gauge block interferometry.

An uncertainty budget is a list of the components, usually associated with an indirect measurement, that contribute to the uncertainty in a measurement result. It identifies, quantifies, and characterizes each independent variable in the measurement equation. It also includes an overall uncertainty analysis incorporating Type A and Type B uncertainties for each variable and the effect of each variable on the measurand. The result is a formal record of an uncertainty analysis that can be used to both characterize and improve the measurement process.

Coordinate Measuring Machine (CMM) Measurements

Product Acceptance Uncertainty: Dimensional Part Inspection with a CMM

This section outlines the uncertainty budget for a spherical mirror measured with a CMM. It describes three types of uncertainty: fitting uncertainty (Type A), process uncertainty (Type A), and standard uncertainty (Type B).

The multivariate statistical methods used to calculate the fitting and process uncertainties of the spherical mirror measurements extend beyond the central methodologies outlined in this book. However, it is included for practitioners that may encounter real multidimensional measurement uncertainty problems requiring both multivariate measurements and analyses. The discussion includes an uncertainty budget (described above) for the spherical mirror measurement performed with a CMM. The measurement is the radius of curvature for a 100 mm diameter mirror.

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Radius of Curvature of a Spherical Mirror

The Radius of Curvature (ROC) of a spherical mirror was measured with a CMM at a manufacturing plant. The nominal dimensions of the mirror are as follows:

1. Radius: 50 mm.
2. Thickness: 12.7 mm.
3. ROC 20 m.

The measurand for this calibration is the mirror's ROC. The uncertainty requirements were $\pm$ 0.125 m ($k = 2$).

An optical image of the mirror mounted for application is shown in Fig. C.1a. Schematics with the nominal dimensions of the mirror are shown in Fig. C.1b and Fig. C.1c.

The Measurement Model

The equation for the radius (squared) of a sphere is

$$R^2 = (x - H)^2 + (y - K)^2 + (z - L)^2, \quad \text{(C.1)}$$

where (x, y, z) are the coordinates at the edge of the sphere, R is the radius, and (H, K, L) are the origin of the sphere. A schematic of a sphere described by Eq. (C.1) is shown in Fig. C.2.

From Eq. (C.1), the maximum change in the z-coordinate (sag) from the center of the mirror to the edge of the mirror is calculated with a nominal ROC of 20 m and an origin of (0, 0, 0):

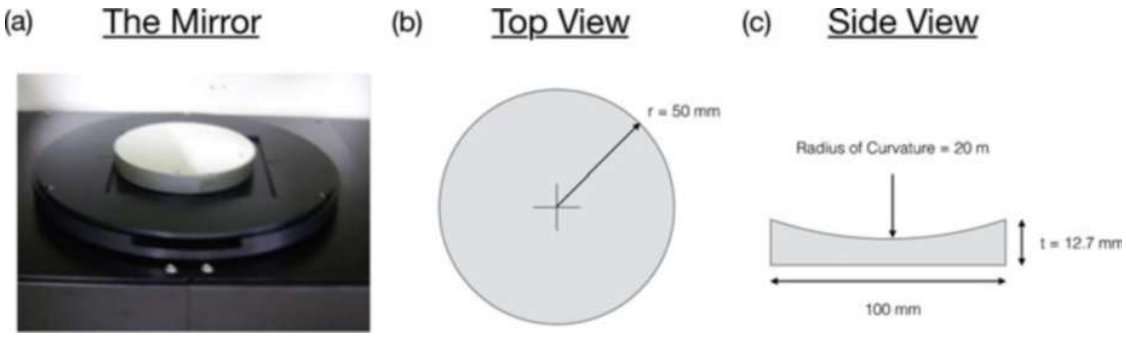
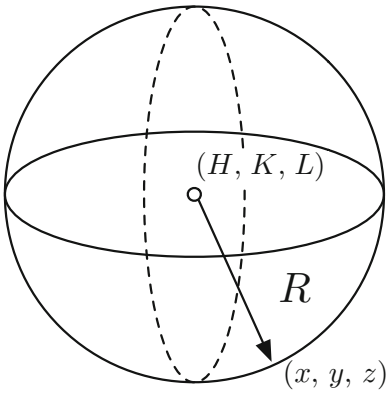


Fig. C.1 (**a**) Optical image, (**b**) top view schematic, and (**c**) side view schematic of the spherical mirror under test



$$R^2 = (x - H)^2 + (y - K)^2 + (z - L)^2$$

Fig. C.2 Equation of a sphere in Cartesian coordinates (x, y, z), with radius R, and origin of (H, K, L)

$$\begin{aligned} z_1 &= (20\ \text{m})^2 - (0\ \text{m})^2 - (0\ \text{m})^2 \\ z_2 &= (20\ \text{m})^2 - (0.05\ \text{m})^2 - (0\ \text{m})^2. \end{aligned} \tag{C.2}$$

The estimation of total sag, Δz:

$$\begin{aligned} \Delta z &= z_1 - z_2 \\ \Delta z &= (20\ \text{m}) - (19.9999375\ \text{m}) \\ \Delta z &= 0.0000625\ \text{m} \\ \Delta z &= 62.5\ \mu\text{m}\ (2460\ \mu\text{in}). \end{aligned} \tag{C.3}$$

With the estimated sag for the spherical mirror of 62.5 μm (2460 μin), measurement considerations should be analyzed to minimize the variations of the z-axis coordinate.

Measurement Considerations

The accuracy in the measurement of the mirror's ROC is dependent on:

1. Surface form of the mirror.
2. Sampling density.
3. CMM probing force.
4. CMM positioning error.

Items 1–3 are discussed below and the CMM positioning error is considered in the uncertainty analysis.

Surface Form of the Mirror

According to the ASME Y14.5 standard (ASME 2018), form tolerances control straightness, flatness circularity, and cylindricity. The form of the mirror is intrinsic to the part itself; however, the mirror mount at the time of measurement could introduce stress and deform the mirror. Line standards measured with a CMM should be mounted on the Airy points to minimize sag at the ends. On the other hand, the Bessel points should be used to minimize 2D sag. For this measurement, the mirror was mounted in its in-use position, i.e., on the mounting chuck for the mirror during use. Based on a degrees of freedom argument (Chap. 11), it was decided that 50 measurement points were adequate for this experiment.

Moreover, measurements near the edge of the mirror are the most sensitive to determining R because the sag, Δz, near the edge of the mirror is the largest. Figure C.3 shows the measurement points used in the calibration of the mirror.

CMM Probing Force

The force used to probe the mirror surface can introduce error in the measurement. Specifically, if the probing force is large, it is possible for probe skidding or probe deflection to occur. Therefore, it is important to make sure the data is collected normal to the surface using a surface normal probing direction. Surface normal probing will minimize the possibility of skidding, leaving only the worry about exceeding the elastic limit of the mirror.

For most glass surfaces, the elastic limit of glass in compression is approximately 50 MPa and is not a major concern. However, care is still taken when probing the surface and an overtravel force of about 0.3 Newtons is used with a backway and trigger of the probe at 0.1 Newtons. Coatings may require some investigation as to

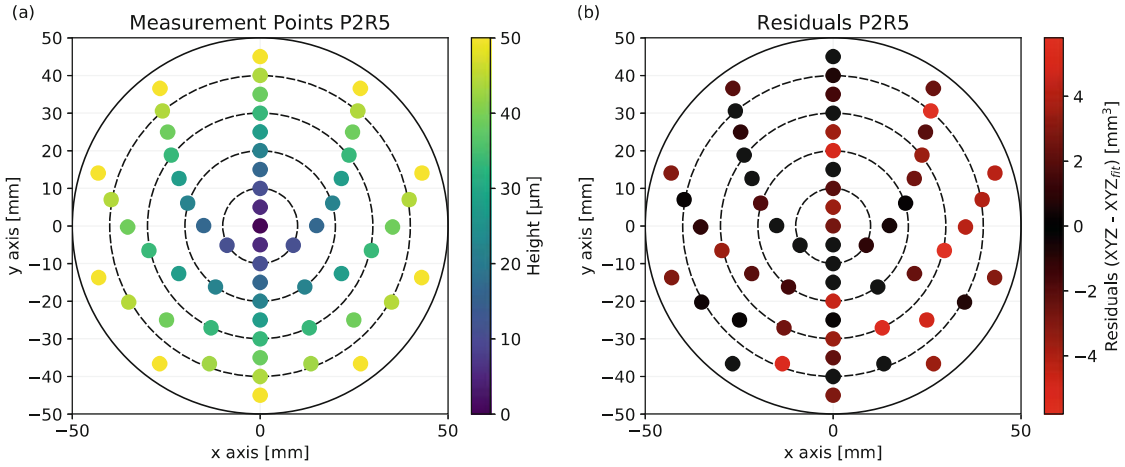


Fig. C.3 (**a**) Location of the 55 measurement points for each run. The measurements of height are taken from Run 5 in Position 2. (**b**) The difference between the measurement data and the measurement model

how they will hold up when touched in this fashion. The probe diameter size can be varied as well. A large probe diameter on a high-quality surface will provide good performance and minimize any surface damage.

The mirror must be stable and not deformed by the fixture method. It must not deflect, move, or bend under triggering forces. Any movement or stresses in fixturing will result in big errors on radius of curvature realized in a free state. Varying the triggering force and extrapolating the evaluated data to zero trigger force usually uncovers some of these problems and gives confidence in the process.

ROC Measurement

The mirror was measured five times at two different locations on the CMM bed, for a total of 10 runs. The temperature and relative humidity during the measurement ranged from 60.08 °F to 68.34 °F and 40.2% RH to 40.6% RH, respectively.

At each position, the operator identified the zero position and then took 5 measurements at the center of the mirror to verify z repeatability and to make sure the probe was set. The operator then took 19 measurements across the surface of the mirror along x-axis, which is the axis that runs perpendicular to the Koba bar, a calibrated step gage (the purpose of the Koba bar is explained in the uncertainty analysis). The operator then took 55 measurements approximately evenly spaced across the surface of the mirror. The operator started approximately 5 mm from the edge of the mirror for a total of 74 measurements per run. Figure C.3a shows the location of the 74 data points taken from Run 5 in mirror Position 2. The color indicates the measured height, which is synonymous to the sag, Δz.

At each measurement point, x, y, and z-coordinates were obtained. The measurement data at each point was transformed to create a new variable so that a linear least-square fit of the data and the measurement model given in Eq. (C.1) could be performed:

$$\left[x^2 + y^2 + z^2\right] = 2xH + 2yK + 2zL - D, \tag{C.4}$$

where H, K, and L are the origin coordinates of the x, y, and z-axis, respectively and D is

$$D = R^2 - H^2 - K^2 - L^2 \tag{C.5}$$

and R is the ROC. A least-squares fit of the data to Eq. (C.4) was performed to extract the parameters H, K, L, and D. As given by Eq. (C.5), the parameter D includes the value for the mirror's ROC. The difference between the measured data and the fit of Eq. (C.2) for Run 5 in mirror Position 2 is shown in Fig. C.3b. The units are in mm^2 because the variable used for the least-squares fit is $x^2 + y^2 + z^2$.

For each of the 10 runs, a mean value for R was extracted, as seen in Table C.1.

Table C.1 Mean value of the ROC for 10 measurements at two locations on the CMM bed

Position	Run	Mean ROC (m)
1	1	20.281
1	2	20.264
1	3	20.294
1	4	20.283
1	5	20.289
2	1	20.272
2	2	20.295
2	3	20.274
2	4	20.274
2	5	20.267

Table C.2 Uncertainty budget table of the ROC measurements of a spherical mirror

Uncertainty component description	Source	Standard uncertainty		Sensitivity coefficients	Type (A or B)	Distribution	Degrees of freedom
CMM process: uncertainty due to process	Process	$s(R_p)$	0.011 μm	1	A	Normal	10
Standards: z-axis repeatability	Process	$s(z)$	0.072 μm	$\frac{\partial R}{\partial z}$	A	Normal	86
Standards: x- and y-axis error	Koba bar	$u(x)$	0.39 μm	$\frac{\partial R}{\partial x}$	B	Normal	59
		$u(y)$	0.39 μm	$\frac{\partial R}{\partial y}$			
Fit: error in H, K, and L	Residuals of fit	$s(H)$	8.1 μm	$\frac{\partial R}{\partial H}$	A	Normal	70
		$s(K)$	6.6 μm	$\frac{\partial R}{\partial K}$			
		$s(L)$	9.5 mm	$\frac{\partial R}{\partial L}$			

Uncertainty Analysis

The uncertainty analysis is organized with an uncertainty budget table (Table C.2). There are four primary contributions to uncertainty: process uncertainty, CMM z-axis positional uncertainty, CMM x-y positional uncertainty, and the uncertainty in fitted values of H, K, L, and D.

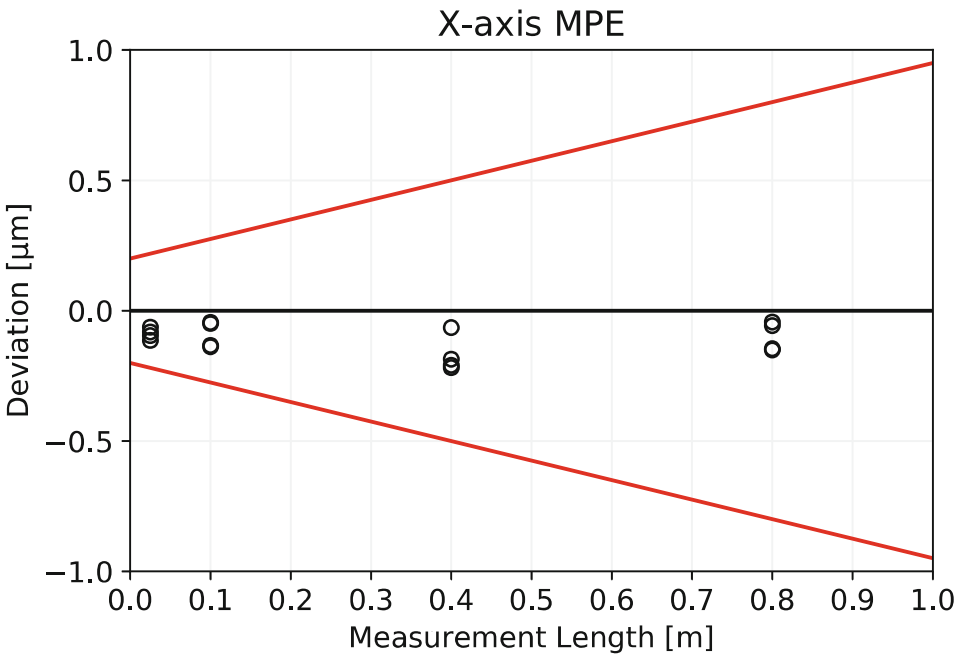


Fig. C.4 Example MPE of a CMM. The manufacturer MPE is plotted with red lines and the measurements in black

CMM Measurement Process Uncertainty

The mirror was placed in two locations on the CMM and the same procedure was run five times at each location. A total of 740 measurements were taken, where each run measured the mirror 74 times. The standard deviation of the mean value for each of the 10 procedure runs was taken to be the standard deviation of the process:

$$s(R_P) = 0.011 \text{ m}. \tag{C.6}$$

CMM Positioning Error (Standards)

The maximum permissible error (MPE), as per ISO 10360-2:2 (ISO 2009), specifies how well a CMM can measure the distance between any two points within the entire CMM measuring volume. The MPE must be valid over the entire CMM volume; therefore, it can overestimate the CMM error within a small measurement volume. For more strict calibrations with lower uncertainty requirements, map the error of the CMM within a small measurement volume. An example MPE plot is shown in Fig. C.4. The manufacturer MPE is plotted in red, while measurements are shown in black.

Due to the strict uncertainty requirements on this measurement, the MPE of the CMM could not be used. Instead, the positioning error of the CMM in the x, y and z-axis were determined at the time of measurement within the measurement volume.

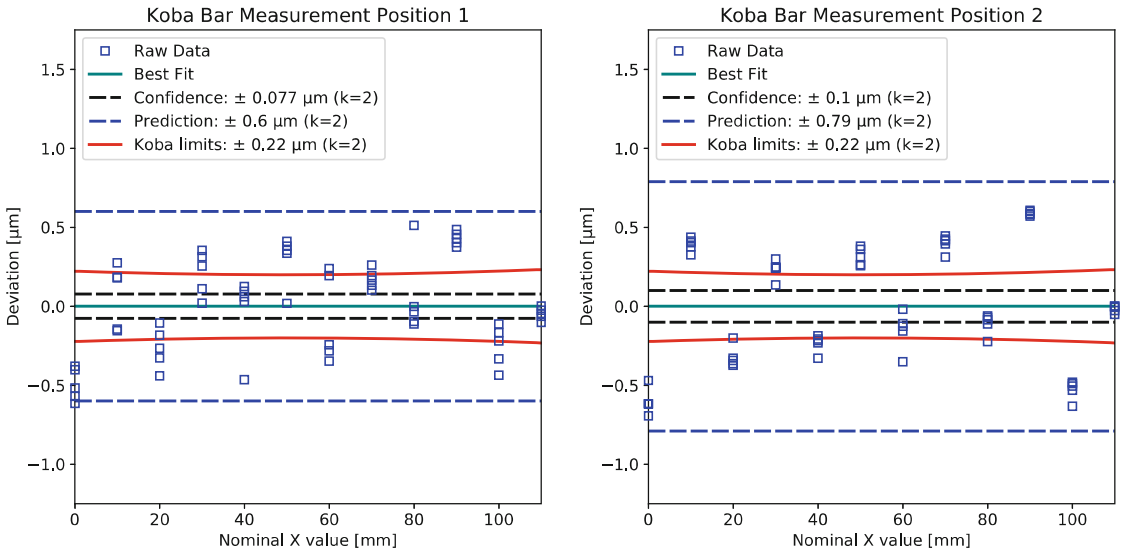


Fig. C.6 Koba bar measurements and best fit, confidence interval, prediction interval, and certified uncertainty

$$s(z) = 0.072\ \mu\text{m}. \tag{C.8}$$

The mirror was placed in two locations on the CMM and the same procedure was run five times at each location. A total of 740 measurements were taken, where each run measured the mirror 74 times. The standard deviation of the mean value for each of the 10 procedure runs was taken to be the standard deviation of the process, $s(R_P)$.

Fit Uncertainty

Type A uncertainty of the fit was determined from the residuals of the fit. Type B uncertainty for numerical error due to the disparity of numbers (20 m radius curvature on a 0.05 m diameter object) was not calculated because the disparity between inputs is within 2 orders of magnitude when the data is organized into the linearized equation. The linearization of the measurement model results in the low process uncertainty and enables CMM measurements on ROC comparable to optical techniques.

Combined Standard Uncertainty: $u(R)$

The final combined standard uncertainty is calculated through an RSS combination of the individual uncertainty components listed in Table C.1 using both the variance and covariance of the parameters and the final computed ROC. The computed standard uncertainty on the ROC due to the z-axis repeatability, x and y-axis error, and the error in H, K, and L is given:

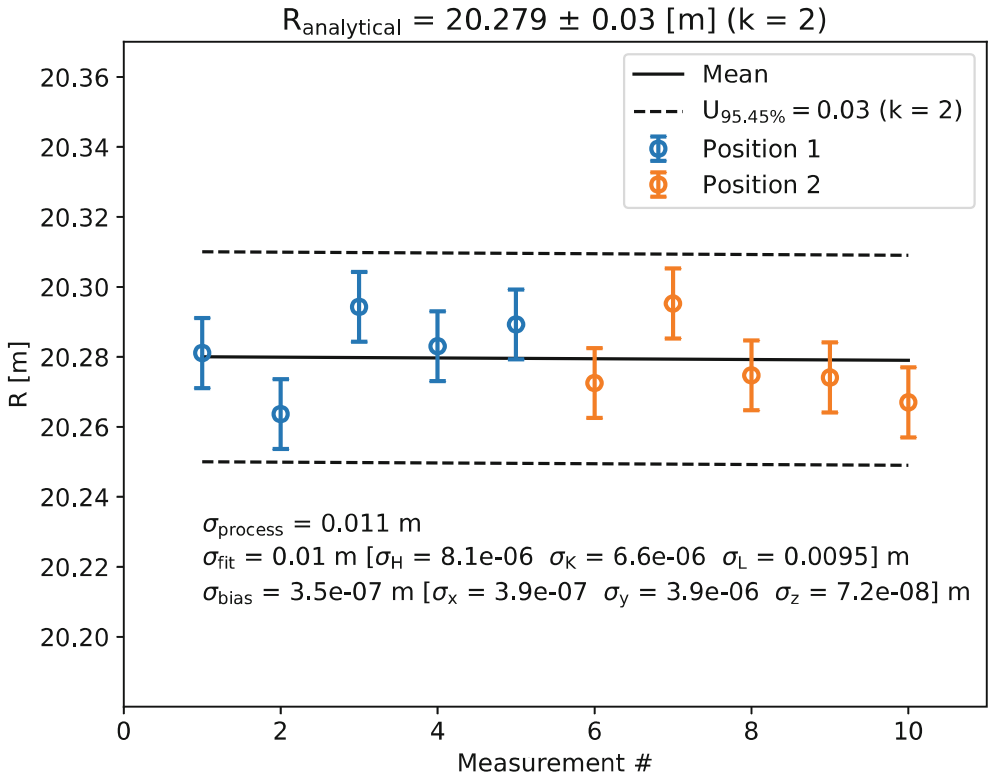


Fig. C.7 Final result on the mirror ROC

$$u(R_o) = f(s(z), u(x), u(y), s(H), s(K), s(L)) = 0.010 \text{ m}. \tag{C.9}$$

The final combined standard uncertainty for the UUT:

$$u^2(R) = \sqrt{u^2(R_o) + u^2(R_p)} = 0.015 \text{ m}. \tag{C.10}$$

With an expansion coefficient ($k = 2$), the final uncertainty: $U_R = 0.030$ m.

Final Results

The final mirror was calibrated with a ROC, $R = 20.279$ m $\pm$ 0.030 m ($k = 2$), which corresponds to a 95.45% confidence interval. The results of all 10 runs are shown in Fig. C.7. A Monte Carlo simulation with $N = 7{,}400{,}000$ was then run to verify the analytical uncertainty results. The data for each run was fit with a random distribution of noise in the x, y, and z-coordinates equal to the uncertainty in each coordinate. The simulation fits the 74 points of a run 10,000 times. There were 10 runs for a total of 7,400,000 points. The results are shown in Fig. C.8 and are consistent with the analytical determination of uncertainty.

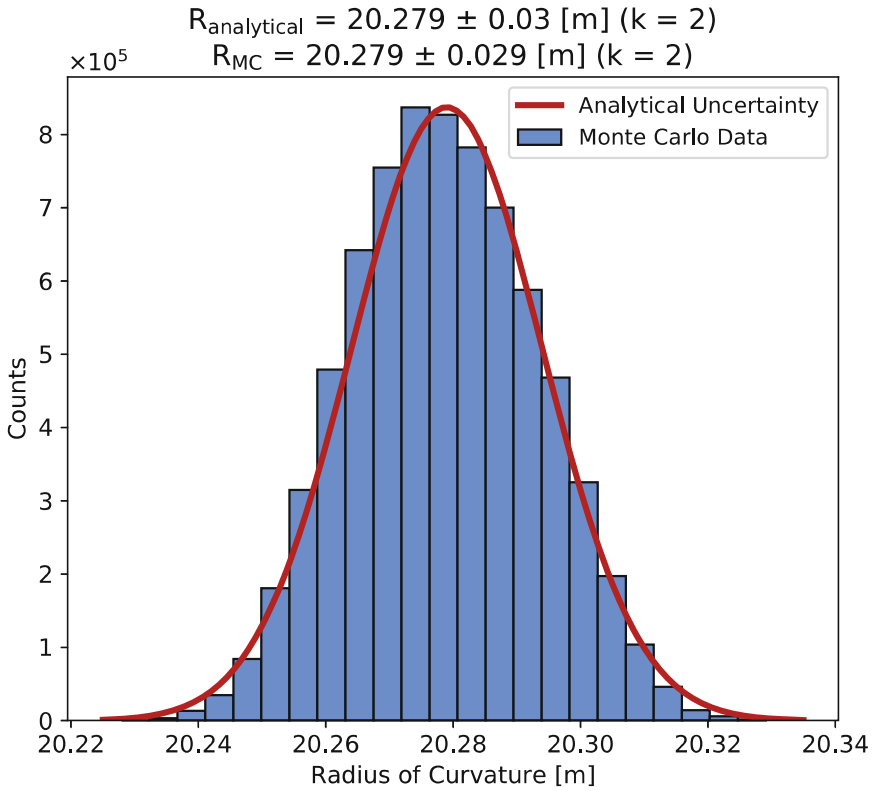


Fig. C.8 Results of Monte Carlo simulations with $N = 7{,}400{,}000$

Related Reading

This Appendix has explored in-depth approaches to uncertainty analysis using a case study in dimensional metrology. The case study itself is not intended to make the reader an expert in these measurements. Uncertainties associated with coordinated measuring machine (CMM) measurements can be especially complex given the dimensional degrees of freedom. ISO 10360-2:2 (ISO 2009) specifies acceptance testing to verify CMM performance and provides guidance on determining maximum permissible error (MPE). An understanding of geometric dimensioning and tolerancing (GD&T) is a requisite for interpreting CMM results, so ASME Y14.5 (ASME 2018) is recommended further reading for anyone using or applying CMM results. Although not discussed here, gage blocks and gage block interferometry form the basis of dimensional measurement traceability. For more detail on gage block interferometry, including development history and the principle of measurement, refer to Ikonen and Riski (1993) and the NBS/NIST publications by Beers (1975) and Doiron and Beers (1995).